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Stochastic Optimization of Spatial Throughput in Wireless Networks under Determinantal Scheduling

Project Context and Motivation

- **Domain**: Wireless communication and stochastic network optimization.

- **Real-world Context:** We are trying to solve the classic network trade-off: packing as many active devices into an area as possible without them drowning each other out.

- **Why is uncertainty intrinsic?**
    - Signals fade unpredictably due to physical environments .
    - Device positions are unpredictable, creating chaotic, constantly changing interference patterns.

Project Objective and Scope Main Goal:

- **Exact Objective:** To transition from the abstract trade-off identified in M1 to concrete mathematical modeling. We aim to deconstruct the network's behavior into definable probability distributions.

- **What is being estimated?**
  - We are not calculating a single "average speed." We are estimating and plotting the exact Probability Density Functions and Cumulative Distribution Functions of the network's underlying variables.

- **Scope for Milestone 2:**
  - To computationally simulate the Fading and geometry to build mathematical intuition before running the full system optimization in Milestone 3.

Project System Overview Step-by-step system:

- **End-to-End System Pipeline:**
  - Stage 1: Generate signal fading matrices.
  - Stage 2: Apply Spatial Point Processes to place nodes.
  - Stage 3: Calculate the stochastic sum of noise.
  - Stage 4: Evaluate SINR to compute Coverage Probability.
- **Data Flow**: 1.)Random Locations + Random Fading
  2.) Aggregate Interference
  3.) SINR
  4.) Success Probability
  5.)Throughput.

Sources of Uncertainty in the Project

- **Measurement Noise:** Multipath propagation causes signal strength to fluctuate wildly over short distances.

- **Environmental Variability:** The exact (x,y) coordinates of transmitting nodes are unknown and constantly changing, creating unpredictable geometric interference.

- **Modeling Uncertainty:** We assume idealized formulas for path loss and interference that may not perfectly capture complex real-world topographies .

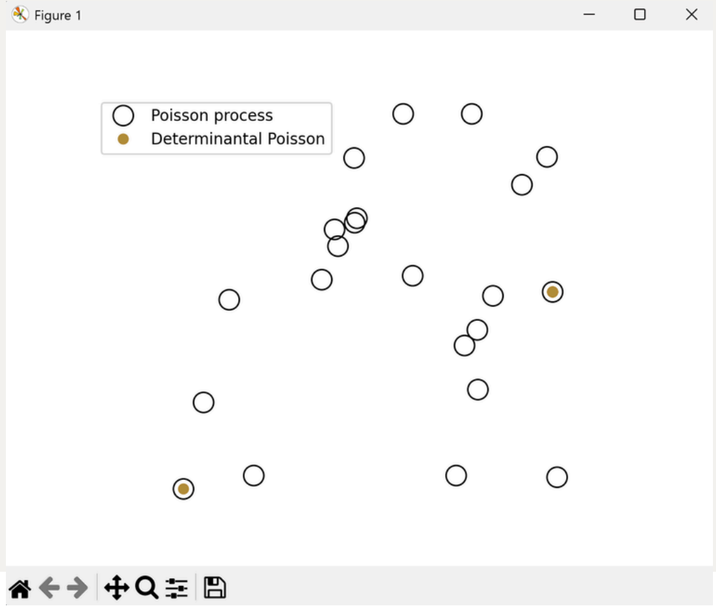
Key Random Variables and Their Roles

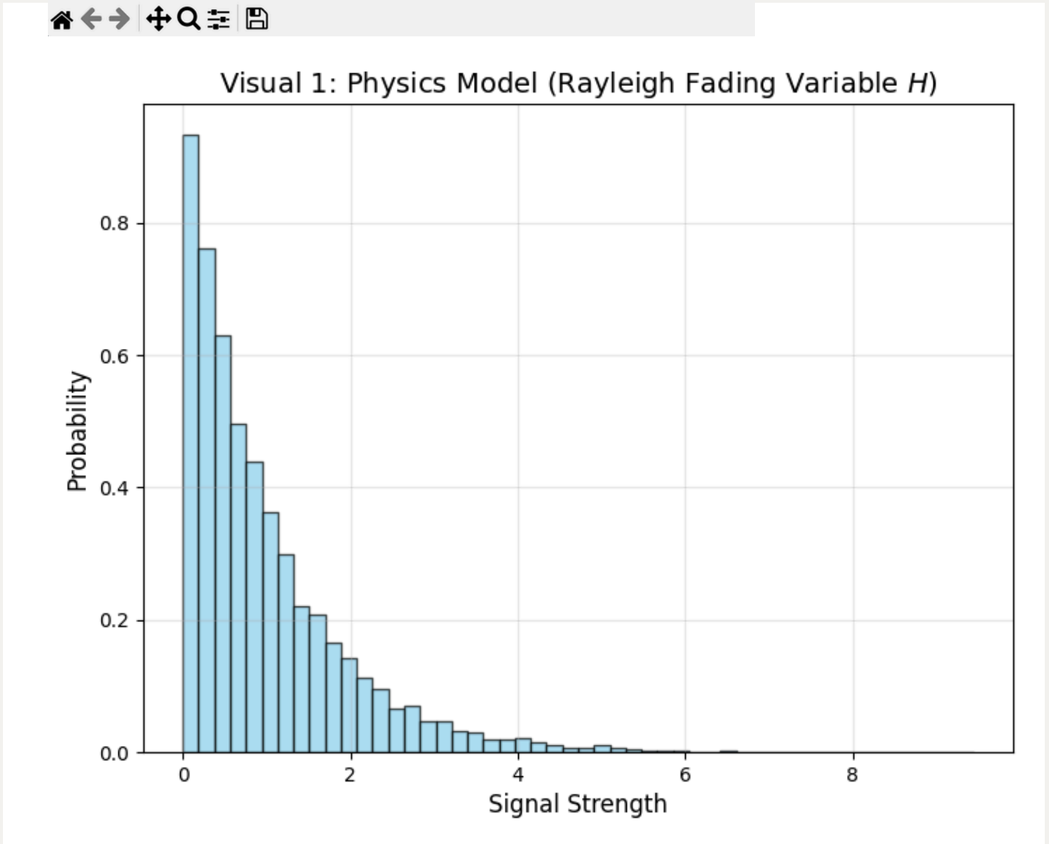
- **Fading Coefficient:** Univariate continuous RV representing random signal strength
- **Node Configuration:** Multivariate spatial point process defining where transmitters exist.
- **Aggregate Interference** : Derived RV representing the total noise from other transmitters.
- **Coverage Probability**: The calculated chance that SINR > threshold.

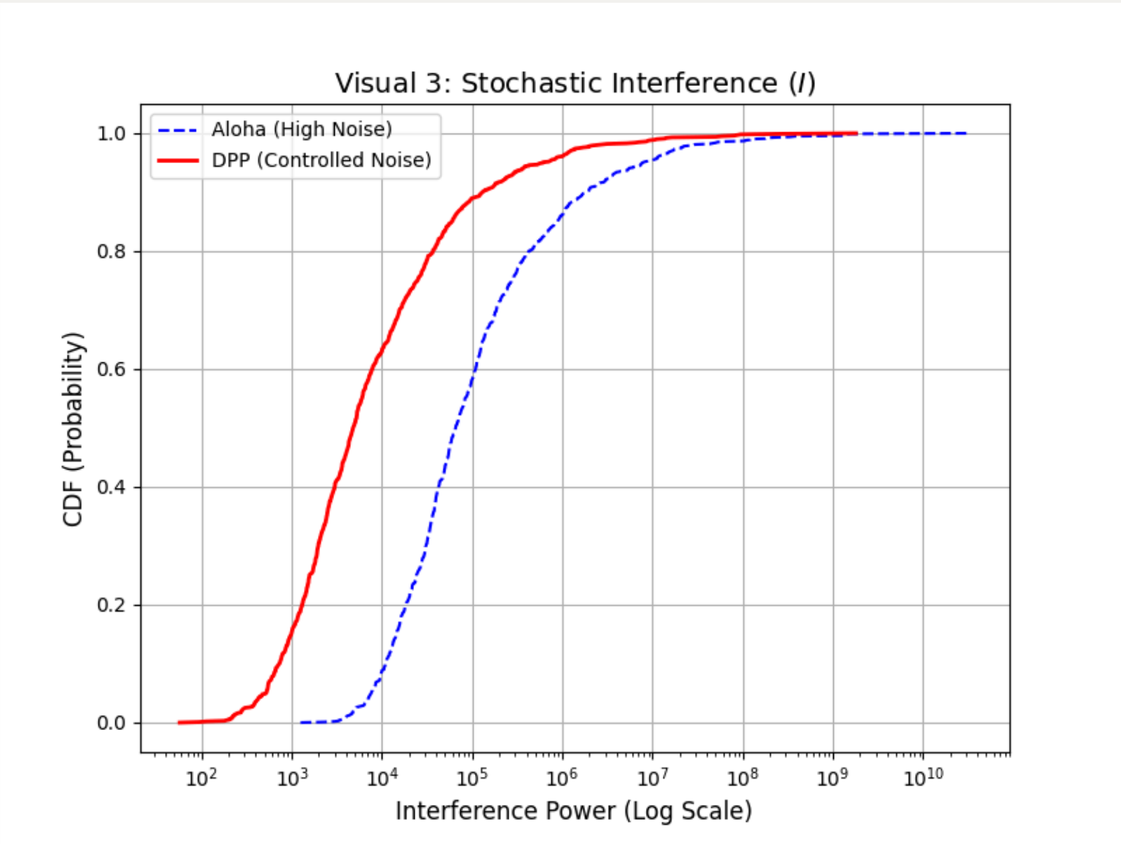
Probabilistic Models and Assumptions

- Fading (H): Exponential Distribution-
  1. Models Rayleigh fading (Non-Line-of-Sight environments).
  2. Captures the memoryless nature of wireless channel power.
- Geometry (Phi): Determinantal Point Process (DPP)
  1. Replaces the baseline Poisson Point Process (PPP / Aloha).
- Dependence Assumptions:
  - PPP (Baseline): Strict independence to random clustering & severe interference spikes.
  - DPP (Proposed): Negative Dependence to Kernel Matrix structurally penalizes proximity and enforces minimum distance constraints.

Graphs And Findings







Probabilistic Reasoning and Inference Logic

- The Foundational Logic Chain: "Random Locations to Path Loss times Fading to Aggregate Interference.

- Step-by-Step Inference Pathway:
  - Implement DPP to introduce Negative Dependence.
  - Structural repulsion physically prevents transmitter clustering.
  - Increased minimum distance limits the extreme upper bounds of Interference (I).
  - Lowered variance and mean of I improves average SINR.
  - Improved SINR unlocks the mathematical headroom for higher network density (lambda).

Operationalizing Probabilistic Reasoning in the Project

How we implement it:

- Use existing Python/MATLAB code
- Change fairness metric to throughput
- Run simulations for different densities

Evaluation Metrics:

- Plotted the Cumulative Distribution Function (CDF) of Aggregate Interference.
- DPP curve shifts significantly left to probabilistic proof of reduced severe noise.
- Comparison between Aloha and DPP

Simplifying Assumptions:

- Ignore edge effects
- Assume fading is independent and identical

Current Limitations and Conceptual Gaps

- Known Weaknesses:
  - Full Buffer Assumption: Assumes 100% continuous data transmission.
  - Yields a "worst-case" interference scenario and overly pessimistic throughput bounds.
- Unverified Assumptions:
  - Assumes ideal statistical independence across all signal fading links.
  - Ignores correlated shadowing (e.g., one large obstacle degrading multiple nearby nodes).
- Conceptual Gaps:
  - Static snapshot methodology lacks a time-domain queuing theory component.
  - Cannot currently predict packet delay or individual user wait times.

Planned Refinements and Role Coordination

Next Steps:

- Implement throughput optimization
- Plot density vs. throughput curves
- Compare peak throughput for DPP and Aloha

Modeling Tasks Ahead:

- Study how interference changes with density
- Analyze the peak point of throughput curves

Summary of Group-Level Understanding

- **Stochastic, Not Deterministic**: Network throughput is an expectation governed entirely by interacting probability distributions.
- **Reshaping Interference:** Shifting from an independent spatial process to a dependent spatial process (DPP) fundamentally reshapes the distribution of aggregate interference.
- **Optimization via Probability:** Enforcing geometric negative dependence is a proven method to maximize network capacity under extreme environmental uncertainty.
- **Project Status:** Mathematical framework verified; prepared to execute full system optimization in Milestone 3.